

REPORT GENERATION AND EXPORT DATA FLOW

Source retrieval, object building, rendering, and month-grouped artifact storage

Scheduling rule: daily always runs; weekly is added on Sunday; monthly is added at month-end when the anchor date is the last day of the month.

1. Bootstrap runtime

Initialize config, MySQL client, repositories, and runtime logging in `main.py`.

2. Resolve periods for this run

Resolve anchor date, report types, and canonical export stems for the batch.

3. Fetch source rows

Pull `energy_kpi`, utility usage, area energy, and selected sensor rows for comparison windows.

4. Aggregate utility sensor statistics

Compute min/avg/max sensor metrics, monitoring structures, and anomaly-ready values.

5. Build domain objects

Create electricity, KPI, and utility blocks with business rules applied before rendering.

6. Assemble unified report_context

Merge meta, sections, flags, labels, notes, and presentation tokens from `report_style.json`.

7. Render and export artifacts

Render `view_html` and `pdf_source_html`, print PDF, and write daily Excel when applicable.

8. Store canonical outputs

Write `output/reports/YYYY_MM/` artifacts with `01_monthly` / `02_weekly` / `03_daily` prefixes.

Period loop

Steps 4–7 repeat for each daily, weekly, or monthly period in the run batch.

Documentation PDF rebuild is a separate workflow and should be triggered only when documentation source files change.